# Venkata Lakshmi Parimala Pasupuleti

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## PROFESSIONAL SUMMARY

Applied AI Engineer with 8+ years of production Python engineering, specializing in **agentic systems, RAG pipelines, and LLM evaluation infrastructure**. Built AgentProbe — an open-source agentic AI testing toolkit with a 5-metric LLM-as-judge eval engine, CI/CD quality gates, ChromaDB RAG observability, and multi-provider LLM routing (Anthropic, OpenAI, Groq, Ollama). Built AI on FHIR — a HIPAA-compliant NLP platform for clinical records, demonstrating production AI in regulated healthcare environments. Shipped Entrata's first production AI pipeline (PyTorch + OpenAI API) across 1.6M real financial records with output validation, quality gates, and compliance controls. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

## WORK EXPERIENCE

### Entrata

Jan 2017 – Aug 2024

**Trainee → Associate Engineer → Software Engineer**

- Pioneered Entrata's **first production AI pipeline** (Python, PyTorch, OpenAI API) across 1.6M real financial records — full lifecycle from research through output validation, quality gates, compliance controls, and production deployment; later productized into a customer-facing feature.
- Achieved **100% PHI compliance** on AWS S3 migration (Libsodium-encrypted pipelines, zero security incidents) — direct analog to healthcare data handling requirements in regulated AI environments.
- Built and owned **20+ production modules** (Python, PostgreSQL, MySQL, JavaScript, RabbitMQ) across 120+ Agile sprints for 100K+ monthly active users and 1M+ property records — 99.5% uptime over 6+ months; cut deployment times 40%, query execution 50%.
- Automated compliance workflows (LIHTC, Section 8/HCV, HUD) with **zero compliance violations** — income eligibility verification, AMI threshold calculations, recertification pipelines across 100K+ unit portfolios.

### George Mason University

Aug 2024 – Aug 2025

**Research Associate — Statistical Modeling · Anomaly Detection**

- Built regression models ($R^2 = 0.977$) in Python and R across 190+ countries; applied Isolation Forest to automate anomaly detection across 3 research pipelines — surfaced anomalies manual review missed.

## PROJECTS

### AgentProbe: Agentic AI Evaluation & Observability Toolkit

Open Source

Python · LangGraph · LangChain · FastAPI · ChromaDB · LLM-as-Judge · Anthropic · OpenAI · Groq · Ollama · React · Docker · GitHub Actions — github.com/venkatalakshmiparimala/AgentProbe

- Built a **5-metric LLM-as-judge eval engine** (factual accuracy, hallucination, retrieval quality, completeness, compliance) with CI/CD quality gates blocking failing agent runs before production — directly mirrors the evaluation infrastructure Pager Health is building for agentic health AI.
- Built a 3-agent LangGraph runtime (Orchestrator, RAG Tracer, Auto Fix) with tool calling, inter-agent state handoffs, and **multi-provider LLM routing** (Anthropic, OpenAI, Groq, Ollama) — 15-endpoint FastAPI backend running 100% locally at zero cost.
- Delivered **trace-level RAG observability** via React dashboard — ChromaDB pipeline tracking per-run metrics, retrieval quality, regression drift, and latency across concurrent agent test suites.

### AI on FHIR: HIPAA-Compliant NLP for Clinical Records

Flask · spaCy · BERT · Transformers · Embeddings · LLMs · React · Docker — github.com/venkatalakshmiparimala/AI_on_FHIR

- Built a clinical NLP platform translating natural language queries to FHIR APIs using spaCy, BERT, and semantic embeddings — **HIPAA-compliant RBAC**, ~60% reduction in retrieval time, bilingual (EN/ES); demonstrates production AI in a regulated healthcare environment directly analogous to Pager Health's domain.

### Leasel: Agentic RAG Pipeline with Pass@k Eval Harness

Python · LangGraph · LLaMA · Transformers · SQLite · reportlab — github.com/venkatalakshmiparimala/Lease_complaince_rag

- Built a production-grade agentic RAG pipeline over 1,240 residential lease PDFs — **93.6% precision, 87.9% F1** across 10 compliance rules; pass@k eval harness (pass@1 81.7%, pass@3 88.3%) with LLM-as-judge scoring of accuracy and reasoning quality; deterministic rules engine keeps decisions fully auditable.

## EDUCATION

**M.S. Data Analytics Engineering** — George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia

**B.Tech Computer Science & Engineering** — Sree Venkateshwara College of Engineering — 2013 – 2017

## TECHNICAL SKILLS

**Languages:** Python, SQL, PL/SQL, JavaScript, TypeScript, R, Bash, HTML/CSS

**Agentic AI & RAG:** LangGraph, LangChain, Multi-Agent Orchestration, Tool Calling, RAG Pipelines, ChromaDB, FAISS, Vector Databases, Pinecone (familiar), Semantic Search, Embeddings, FastAPI, REST APIs

**LLM & Eval:** LLM-as-Judge, LLM Evaluation, Eval Frameworks, Pass@k Harnesses, Hallucination Detection, CI/CD Quality Gates, Prompt Engineering, OpenAI API, Anthropic API, Groq, Ollama, AWS Bedrock (familiar), Azure OpenAI (familiar)

**ML & NLP:** PyTorch, Scikit-learn, spaCy, BERT, Transformers, XGBoost, Pandas, NumPy, Hugging Face, Feature Engineering

**Cloud & Infrastructure:** AWS (S3, EC2, RDS, Glue), Docker, CI/CD (GitHub Actions), PostgreSQL, MySQL, SQLite, Git, Linux, HIPAA Compliance, Agile/Scrum (120+ sprints)